

Trainings ▾

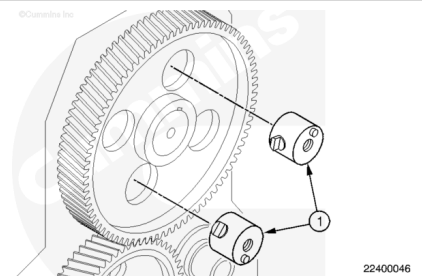
Preparatory Steps

- Remove the gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-031-tr.html>)
- Remove the camshaft idler gear. Refer to Procedure 001-036 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-036-tr.html>)

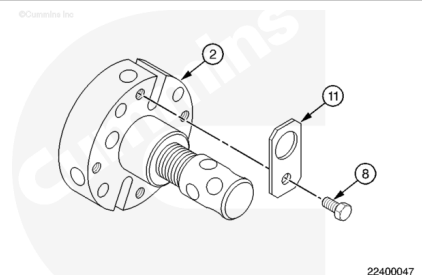


Remove

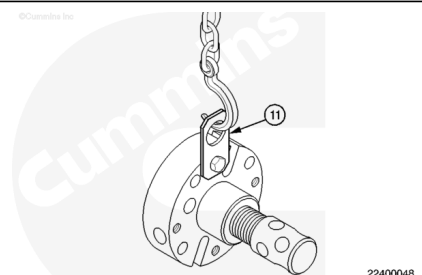
Place two puller adapter assemblies (1) in opposite lightening holes in cam gear.



Install lifting plate (11) on the holder plate (2) with one screw (8). Tighten screw securely.

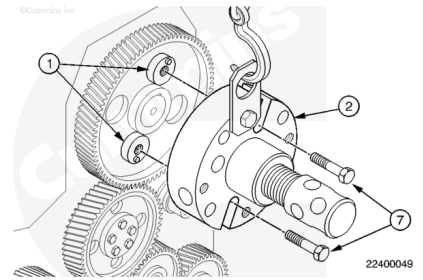


Attach a lifting device to the lifting plate (11).



⚠ WARNING ⚠

The puller assembly is heavy. Use a lifting device when securing it to the cam gear.



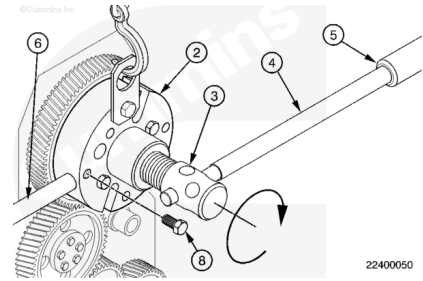
Make sure the forcing screw pilot is engaged with the center of the camshaft and the puller adapter assembly studs are properly seated in the holder plate assembly.

Using the lifting device, position the holder plate (2) against the cam gear and install two screws (7) through the holder plate (2) into the puller adapters (1).

Tighten screws securely.

⚠ WARNING ⚠

Cam gear and puller assembly may fall when released and cause serious personal injury.



Do not attempt to alter the puller assembly for use with any kind of impact wrench. The threads on the puller assembly may gall and seize.

Use the lever bar (4), extension bar (5), and torque reaction lever (6), to turn the forcing screw (3) clockwise, pulling the cam gear from the camshaft.

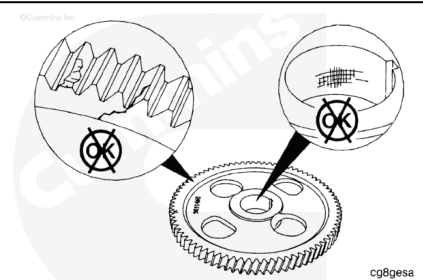
Inspect for Reuse

Inspect the camshaft gear for cracks and chipped or broken teeth.

Inspect the bore of the gear for frets or burrs.

Remove any burrs with Scotch-Brite™ 7448, Part Number 3823258, and a fine honing stone or equivalent.

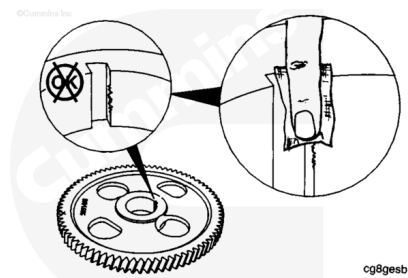
If the frets, burrs, or raised material can **not** be removed, replace the gear.



Inspect the gear keyway for burrs.

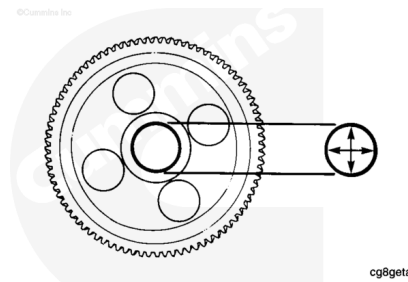
Remove burrs with Scotch-Brite™ 7448, Part Number 3823258, and a fine honing stone or equivalent.

If the keyway is damaged or the burrs can **not** be removed, the gear **must** be replaced.



Measure the camshaft gear bore inside diameter.

Camshaft Gear Bore I.D.		
mm		in
46.912	MIN	1.8469
46.938	MAX	1.8479

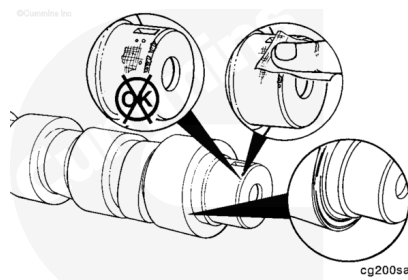


Inspect the camshaft nose for frets or burrs.

Remove any burrs with Scotch-Brite™ 7448, Part Number 3823258, and a fine honing stone or equivalent.

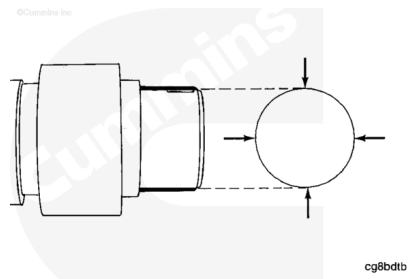
If frets or burrs can **not** be removed, replace the camshaft.

Inspect the thrust surface of the camshaft for grooves or heavy wear.



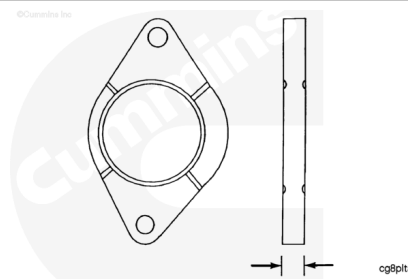
Measure the camshaft gear journal outside diameter.

Camshaft Gear Journal O.D.		
mm		in
46.987	MIN	1.8499
47.013	MAX	1.8509



Measure the camshaft thrust plate thickness.

Camshaft Thrust Plate Thickness		
mm		in
8.96	MIN	0.353
9.04	MAX	0.356

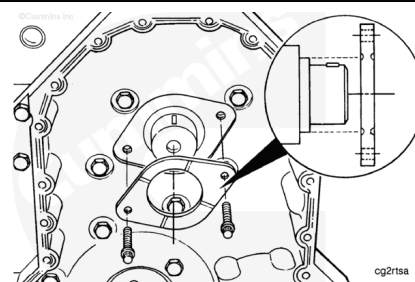


Install

Install the thrust plate on the camshaft.

Install the capscrews and tighten.

Torque Value: 47 n•m [35 ft-lb]

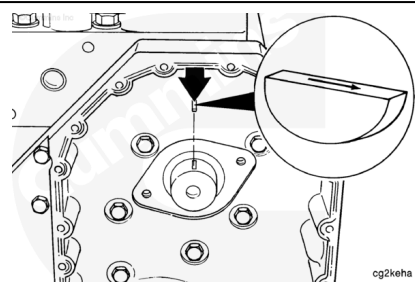


Gear housing clearance makes key installation more difficult.
Use care in seating the key.

Be sure the gear mounting surface is clean and free of oil, dirt, or debris.

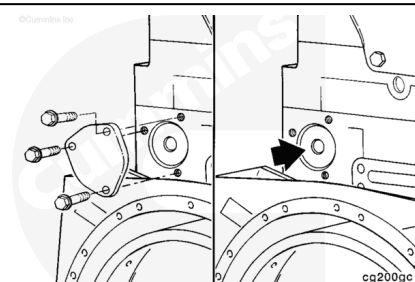
If the camshaft gear is used again, use the same part number key as the one that was removed. Be sure the arrow on the key is pointing in the same direction as when it was removed.

Use a leather hammer to install the camshaft gear key.



It can be necessary to remove the camshaft rear cover plate and apply pressure to the end of the camshaft to hold it in the forward position while the gear is being installed. Refer to Procedure 001-008 in Section 1.

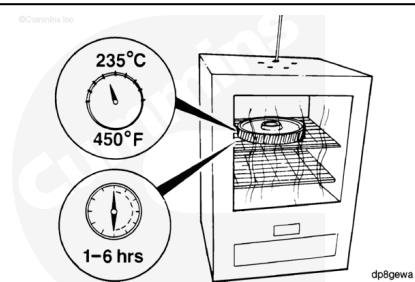
(/qs3/pubsys2/xml/en/procedures/35/35-001-008-tr.html)



Heat the gear in an oven for a minimum of 1 hour, but **not** longer than 6 hours.

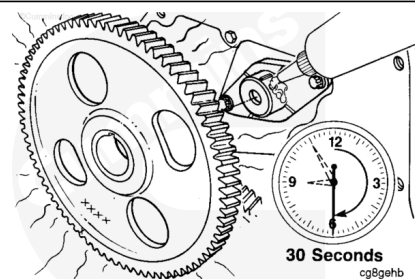
Measurements

	celsius	fahrenheit
Temperature:	235	450



⚠ WARNING ⚠

Use insulated gloves, Part Number 3823730, and/or hot clamp pliers, Part Number 3823732, when handling heated parts. Hot parts can cause serious personal injury.



⚠ CAUTION ⚠

The timing marks and gear part number must be facing away from the camshaft when the gear is installed to prevent engine damage.

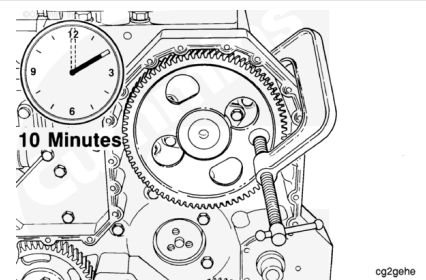
Use Lubriplate™ 105 or equivalent, to coat the camshaft nose before installing the camshaft gear.

Remove the gear from the oven. Install the gear on the camshaft within 30 seconds after removing from the oven.

Align the gear keyway with the key in the camshaft, and install the gear on the camshaft.

Install a C-clamp around the gear housing and the camshaft gear to hold the gear in position as it cools.

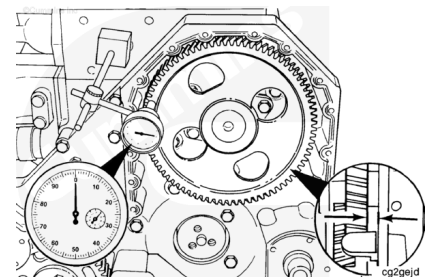
Remove the C-clamp after the gear has cooled to room temperature (approximately 10 minutes).



Use a dial indicator to check the end clearance of the gear.

Push the camshaft to the rear and set the gauge at zero.

Pull the gear out and measure the end clearance.



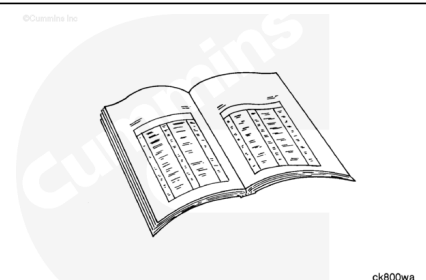
Camshaft Gear End Clearance

mm		in
0.13	MIN	0.005
0.41	MAX	0.016

Finishing Steps

⚠ CAUTION ⚠

Failure to check the engine timing can cause severe engine damage.



- Install the camshaft idler gear. Refer to Procedure 001-036 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-036-tr.html>)
- Check the engine timing. Refer to Procedure 006-025 in Section 6. (</qs3/pubsys2/xml/en/procedures/35/35-006-025-tr.html>)
- Install the gear cover. Refer to Procedure 001-031 in Section 1. (</qs3/pubsys2/xml/en/procedures/35/35-001-031-tr.html>)
- Operate the engine to normal operating temperature and check for leaks.